

2.2 Group 2

Question Paper

Course	CIEA Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.2 Group 2
Difficulty	Medium

Time allowed: 70

Score: /54

Percentage: /100

Question 1a

Magnesium, calcium and barium are some of the Group 2 elements of the Periodic Table.

State and explain the trend in the atomic radius of the Group 2 elements from beryllium to strontium.

[2 marks]

Question 1b

Magnesium is frequently used in fireworks. It burns rapidly in oxygen, and it adds sparks which generally improves the appearance and brilliance of the firework.

Write an equation to represent the reaction which takes place when a strip of magnesium is burned in oxygen and state an observation which would be made.

State symbols are **not** required.

[2 marks]

Question 1c

Group 2 compounds, such as the hydroxides of magnesium and calcium, have a number of other uses.

i)

Write equations to show the formation of magnesium hydroxide and calcium hydroxide from their metals.

ii)

Give an approximate pH value and a use for each of the hydroxides.

[5 marks]

Question 1d

Group 2 elements will react with water.

A student added 20 cm^3 of water to two boiling tubes. A small piece of calcium was added to one of the boiling tubes and a small piece of barium was added to the other.

State **two** observations that the student would make and how the observations would differ between the two boiling tubes.

[3 marks]

Question 2a

Group 2 hydroxides, $\text{M}(\text{OH})_2$, can be used to neutralise acids.

Write an equation for the reaction of barium with cold water.

State symbols are **not** required.

[1 mark]

Question 2b

Explain why a saturated solution of barium hydroxide is more alkaline than a saturated solution of magnesium hydroxide.

[2 marks]

Question 2c

An unknown solution of a group 2 metal salt was tested for the presence of carbonate ions.

Dilute hydrochloric acid was added to the solution and the gas given off was bubbled through limewater. A white precipitate formed.

i)

Write the **ionic** equation for the formation of the gas. State symbols are **not** required.

[1]

ii)

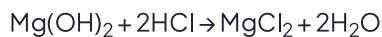
Write the equation for the formation of the precipitate. Include state symbols.

[2]

[3 marks]

Question 2d

Magnesium hydroxide can be used as an indigestion remedy to neutralise hydrochloric acid in the stomach.



Calculate the minimum mass of magnesium hydroxide needed to neutralise 0.210 mol of hydrochloric acid.

Give your answer to an appropriate number of significant figures.

[3 marks]

Question 3a

Group 2 metals and their compounds are used commercially in a variety of processes and applications.

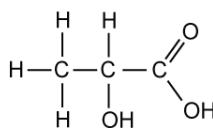
Calcium carbonate is an insoluble solid that can be used to lower the acidity of lake water.

Explain why the rate of this reaction decreases when the water temperature falls.

[3 marks]

Question 3b

Calcium carbonate reacts with lactic acid (Figure 3.1) to form calcium lactate, which is used to treat or prevent low blood calcium levels (hypocalcemia).



Lactic acid

Figure 3.1

State the **two** other products of the reaction of calcium carbonate with lactic acid.

[1 mark]

Question 3c

Magnesium can be used to extract titanium.

i)

Write an equation to show how magnesium is used as the reducing agent in the reaction of magnesium with titanium(IV) chloride. State symbols are **not** required.

(2)

ii)

Explain, in terms of oxidation numbers, why magnesium is the reducing agent.

(1)

iii)

The excess magnesium used in this extraction is removed by reacting it with dilute sulfuric acid to form magnesium sulfate.

Explain why the magnesium sulfate formed is easy to separate from the titanium.

(1)

[4 marks]

Question 3d

Strontium metal is used in the manufacture of alloys.

Explain why strontium has a higher melting point than barium.

[2 marks]

Question 4a

Calcium salts have a number of industrial, agricultural and medicinal uses.

Calcium carbonate is used as an antacid to relieve heartburn, acid indigestion, and upset stomach. Calcium oxide is used to make porcelain, glass, mortars and cement. Calcium hydroxide is used in agriculture when a rapid change in the pH of soil is required.

These three calcium salts are connected by the Lime cycle shown in Fig. 4.1.

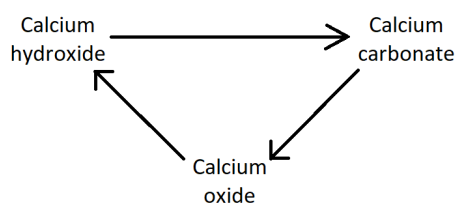


Fig. 4.1

Write **three** balanced chemical equations to demonstrate the reactions that occur in the Lime cycle. Include state symbols.

[4 marks]

Question 4b

In agriculture, bulk lime spreaders are used to cover fields with solid calcium salts, including calcium carbonate, calcium oxide and calcium hydroxide.

i)

Explain how solid calcium hydroxide increases the pH of acidic soil.

[3]

ii)

Write an **ionic** equation to show the reaction that causes the pH of the soil to increase. State symbols are **not** required.

[1]

[4 marks]

Question 4c

When calcium carbonate is heated strongly it undergoes a process called calcination.

i)

What is the chemical reaction that is described by calcination?

[1]

ii)

Explain why strontium carbonate has greater thermal stability than calcium carbonate.

[3]

[4 marks]

Question 4d

A sample of hydrated calcium nitrate crystals was gently heated in a test tube. With further heating, a mixture of two gases was evolved. One of the gases was coloured and the other was colourless.

- i)
Identify the coloured gas and give its colour.

[1]

- ii)
Identify and describe a test, including the result, for the colourless gas.

[1]

[2 marks]

Question 5a

This question is about Group 2 elements and their compounds.

Explain why the first ionisation energy of calcium (590 kJ mol^{-1}) is greater than that of strontium (550 kJ mol^{-1}).

[2 marks]

Question 5b

Strontium chloride is used in toothpaste for sensitive teeth.

i)

Write a balanced chemical equation to show the formation of strontium chloride from strontium and hydrochloric acid. Include state symbols.

[2]

ii)

Suggest why this reaction is not used to produce strontium chloride on an industrial scale.

[1]

[3 marks]

Question 5c

Magnesium sulfate solution can be given as first aid to someone who has accidentally swallowed barium chloride.

Explain, with the help of an **ionic** equation, why drinking magnesium sulfate solution is an effective treatment for barium poisoning. State symbols are **not** required.

[2 marks]

Question 5d

Milk of magnesia, $\text{Mg}(\text{OH})_2$, and calcium carbonate can both be used in the effective relief of acid indigestion.

By comparing the chemical equations for their reactions with hydrochloric acid, justify which remedy might cause the person taking it to have wind.

[2 marks]

Model Answers: Medium

1a

a) The trend in the atomic radius of Group 2 elements from beryllium to strontium is:

- The atomic radius increases / atoms get bigger (as you descend Group 2); [1 mark]
- (Because) there are more shells / energy levels

OR

The number of shells / energy levels increases; [1 mark]

[Total: 2 marks]

- This is a Group 2 trend that you need to learn and be able to explain
- As you descend the group:
 - Each atom has the same effective nuclear charge of +2
 - Each atom gains an additional shell of electrons
 - This means that the force of attraction between the nucleus and the outermost electrons becomes weaker resulting in the atomic radius increasing

1b

b) When magnesium burns in oxygen:

- $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$; [1 mark]
- Magnesium burns with a bright white light; [1 mark]

[Total: 2 marks]

- Burning magnesium in oxygen is a common demonstration; you need to be able to state the observation which would be made
 - The bright white light is dangerous if looked at directly – it can cause temporary loss of sight
 - A blue screen / blue glass must be used
- The burning of magnesium also generates intense heat which can cause burns or could initiate combustion of any flammable materials which are too close

1c

c)

i) Equations for the formation of calcium hydroxide and magnesium hydroxide from magnesium and calcium as elemental metals are:

- $\text{Mg} + 2\text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2 + \text{H}_2$; [1 mark]
- $\text{Ca} + 2\text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2 + \text{H}_2$; [1 mark]

ii) The pH values and uses of calcium hydroxide and magnesium hydroxide are:

- pH value of both hydroxides must be between 8 and 14

AND

The pH value of calcium hydroxide must be higher than the pH value stated for magnesium hydroxide; [1 mark]

- Magnesium hydroxide is used as a cure / remedy for indigestion

OR

Magnesium hydroxide is milk of magnesia; [1 mark]

- Calcium hydroxide is used in agriculture

OR

Calcium hydroxide is used to neutralise / raise the pH of acidic soil; [1 mark]

[Total: 5 marks]

- As well as the trends of Group 2, you need to know the reactions which the Group 2 elements undergo and the uses of the Group 2 compounds
- You should be able to provide approximate pH values for the hydroxides of Group 2

1d

d) Comparing the reaction of calcium with water and barium with water:

Observations:

- The student would see effervescence / bubbles / fizzing in both boiling tubes

OR

The student would see effervescence / bubbles / fizzing with calcium **AND** barium; [1 mark]

- The solid would dissolve / disappear in both boiling tubes; [1 mark]

Difference:

- The reaction would be more vigorous / violent with barium

OR

The solid would dissolve / disappear faster with barium

OR

A colourless solution would form with barium; [1 mark]

[Total: 3 marks]

- The reactions of Group 2 elements with water are asked about very often in exam questions
 - You must be able to describe the reactions of different Group 2 metals with water, comparing them to each other and must be able to provide reaction equations
- The reactivity of Group 2 elements increases down the group, so the reaction of barium with water is more vigorous than of calcium with water
 - The atoms get bigger and the outermost electrons become further from the nucleus
 - The attraction between the nucleus and the outermost electrons is weaker as you descend the group, meaning the outermost electrons are lost more easily
 - This makes the elements lower down the group more reactive

2a

a) The equation for the reaction of barium with cold water is:

- $\text{Ba} + 2\text{H}_2\text{O} \rightarrow \text{Ba}(\text{OH})_2 + \text{H}_2$; [1 mark]

[Total: 1 mark]

- Barium reacts with cold water to form an alkaline solution of barium hydroxide and hydrogen
 - $\text{Ba} + \text{H}_2\text{O} \rightarrow \text{Ba}(\text{OH})_2 + \text{H}_2$
- The oxygen and hydrogen atoms are unbalanced
 - 1 oxygen atom on the left-hand side and 2 on the right
 - 2 hydrogen atoms on the left-hand side and 4 on the right
 - This can be balanced by having $2\text{H}_2\text{O}$ on the left-hand side
 - $\text{Ba} + 2\text{H}_2\text{O} \rightarrow \text{Ba}(\text{OH})_2 + \text{H}_2$

2b

b) A saturated solution of barium hydroxide is more alkaline than a saturated solution of magnesium hydroxide because:

- Barium hydroxide is more soluble than magnesium hydroxide

OR

The solubility of the hydroxides increases going down Group 2; [1 mark]

- The concentration of hydroxide / OH^- ions is greater (in barium hydroxide)

OR

There are more hydroxide / OH^- ions in (barium hydroxide) solution; [1 mark]

[Total: 2 marks]

- The Group 2 metal hydroxides can dissolve into solution forming the M^{2+} ion and 2OH^- ions
- The solubility of the Group 2 metal hydroxides increases as you move down the group
- This means that as you move down the group, the concentration of OH^- ions increases
- Therefore, the pH increases / the solution becomes more alkaline

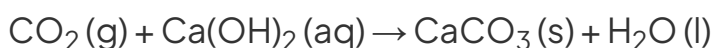
2c

c)

i) The **ionic** equation for the formation of the gas is:

- $\text{CO}_3^{2-} + 2\text{H}^+ \rightarrow \text{CO}_2 + \text{H}_2\text{O}$; [1 mark]

ii) The equation for the formation of the precipitate is:



- Correct balanced equation; [1 mark]
- Correct state symbols; [1 mark]

[Total: 3 marks]

- For part (i)
 - The carbon dioxide gas is formed by the reaction of the metal carbonate with

hydrochloric acid

- $\text{MCO}_3 + 2\text{HCl} \rightarrow \text{MCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$
- You then need to convert this into the ionic equation
 - $\text{M}^{2+} + \text{CO}_3^{2-} + 2\text{H}^+ + 2\text{Cl}^- \rightarrow \text{M}^{2+} + 2\text{Cl}^- + \text{CO}_2 + \text{H}_2\text{O}$
- You then remove the spectator ions
 - $\text{CO}_3^{2-} + 2\text{H}^+ \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
- For part (ii)
 - The precipitate formed by bubbling carbon dioxide through limewater is solid calcium carbonate / $\text{CaCO}_3(\text{s})$
 - You are expected to know that limewater is calcium hydroxide solution / $\text{Ca}(\text{OH})_2(\text{aq})$
 - Carbon dioxide is an acidic gas, so this is a neutralisation reaction
 - Acid + base $\rightarrow$ salt + water
 - $\text{CO}_2(\text{g}) + \text{Ca}(\text{OH})_2(\text{aq}) \rightarrow \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$
 - There is no balancing required for this equation

[Total: 3 marks]

- For part (i)
 - The carbon dioxide gas is formed by the reaction of the metal carbonate with hydrochloric acid
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 - You then need to convert this into the ionic equation
 - $\text{M}^{2+} + \text{CO}_3^{2-} + 2\text{H}^+ + 2\text{Cl}^- \rightarrow \text{M}^{2+} + 2\text{Cl}^- + \text{CO}_2 + \text{H}_2\text{O}$
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 - There is no balancing required for this equation

2d

d) To calculate the minimum mass of magnesium hydroxide needed to neutralise 0.210 mol of hydrochloric acid:

- Moles of $\text{Mg}(\text{OH})_2 = 0.105$; [1 mark]
- M_r of $\text{Mg}(\text{OH})_2 = 58.3$; [1 mark]
- Mass of $\text{Mg}(\text{OH})_2 = 6.1215(\text{g})$

AND

Answer given to 2 significant figures = 6.1 (g) **OR** to 3 significant figures = 6.12 (g); [1 mark]

[Total: 3 marks]

- There is a 1 : 2 stoichiometric relationship between the $\text{Mg}(\text{OH})_2$ and HCl
 - So, the number of moles of $\text{Mg}(\text{OH})_2 = \frac{0.210}{2} = 0.105$ moles
- The M_r of $\text{Mg}(\text{OH})_2$ is needed to calculate the minimum mass
 - M_r of $\text{Mg}(\text{OH})_2 = 24.3 + (2 \times (16.0 + 1.0)) = 58.3$
- Mass = moles $\times M_r$

- So, the mass of $\text{Mg}(\text{OH})_2 = 0.105 \times 58.3 = 6.1215 \text{ g}$
- **Remember:** The question asks for your answer to the appropriate number of significant figures, which for this question is 2 or 3 significant figures

3a

a) The rate of this reaction decreases when the water temperature falls because:

- A decrease in temperature decreases the energy of the particles / ions / H^+ (in the lake); [1 mark]
- Which decreases the number of particles / ions / H^+ (in the lake) with $E \geq E_a$

OR

Which decreases the number of particles / ions / H^+ (in the lake) with sufficient energy to react; [1 mark]

- So there are less frequent successful / effective / productive collisions; [1 mark]

[Total: 3 marks]

- Decreasing temperature results in the particles having less kinetic energy
- This means that more particles don't have the required energy to react, i.e. $E < E_a$
- Therefore, there are less frequent successful collisions / less successful collisions per unit time, which results in a slower rate of reaction

3b

b) The **two** other products of the reaction of calcium carbonate with lactic acid are:

- Carbon dioxide / CO_2

AND

Water / H_2O ; [1 mark]

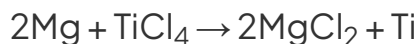
[Total: 1 mark]

- The displayed formula of lactic acid shows that it is a carboxylic acid
- This means that the question is asking for the products of a neutralisation reaction between a metal carbonate and a carboxylic acid
 - Metal carbonate + carboxylic acid $\rightarrow$ metal salt + carbon dioxide + water

3c

c)

i) The equation to show how magnesium is used as the reducing agent in the reaction of magnesium with titanium(IV) chloride is:



- Correct chemical species; [1 mark]
- Correct balancing; [1 mark]

ii) In terms of oxidation states, magnesium is the reducing agent because:

- Mg changes its oxidation number from 0 to +2, so electrons are lost
OR
Ti changes its oxidation number from +4 to 0, so electrons are gained; [1 mark]

iii) The magnesium sulfate formed is easy to separate from the titanium because:

- Magnesium sulfate / it is soluble in water
AND
So, the titanium can be filtered from the solution; [1 mark]

[Total: 4 marks]

- For part (i)
 - You are told that the magnesium reacts with titanium(IV) chloride to produce titanium
 - $\text{Mg} + \text{TiCl}_4 \rightarrow \text{Ti}$
 - The other product must be magnesium chloride
 - $\text{Mg} + \text{TiCl}_4 \rightarrow \text{MgCl}_2 + \text{Ti}$
 - There are four chlorine atoms on the left-hand side and two on the right, so you need 2MgCl_2
 - $\text{Mg} + \text{TiCl}_4 \rightarrow 2\text{MgCl}_2 + \text{Ti}$
 - There are now two magnesium atoms on the right-hand side and one on the left, so you need 2Mg
 - $2\text{Mg} + \text{TiCl}_4 \rightarrow 2\text{MgCl}_2 + \text{Ti}$
- For part (ii)
 - **Remember:** Reducing agents cause reduction in other chemicals and are themselves oxidised
 - The oxidation number of the magnesium starts at 0 in Mg and becomes +2 in MgCl_2

- This increase in the oxidation number is oxidation, which means that magnesium is the reducing agent
- For part (iii)
 - Magnesium sulfate is soluble in water
 - Titanium metal is solid and does not dissolve in / react with water
 - This means that the titanium can be filtered from the mixture

3d

d) Strontium has a higher melting point than barium because:

- The delocalised electrons (in strontium) are closer to the positive ions / cations / atom / nuclei

OR

The positive ions / cations / atoms (in strontium) are smaller

OR

The positive ions / cations / atoms have fewer (electron) shells / levels; [1 mark]

- So, strontium has a stronger attraction between the positive ions / cations / atoms / nuclei and the delocalised electrons

OR

So, strontium has stronger metallic bonding; [1 mark]

[Total: 2 marks]

- As you move down Group 2:
 - The elements have the same equivalent nuclear charge of +2
 - The elements have increased shielding
 - This results in a weaker force of attraction between the nuclei and the outermost electrons
- For the metallic bonding present in Group 2 metals, this means that the force of attraction / metallic bond gets weaker as you move down the group

4a

a) The **three** balanced chemical equations for the Lime cycle are:

- $\text{Ca(OH)}_2 (\text{aq}) + \text{CO}_2 (\text{g}) \rightarrow \text{CaCO}_3 (\text{s}) + \text{H}_2\text{O} (\text{l});$ [1 mark]
- $\text{CaCO}_3 (\text{s}) \rightarrow \text{CaO} (\text{s}) + \text{CO}_2 (\text{g});$ [1 mark]

- $\text{CaO (s)} + \text{H}_2\text{O (l)} \rightarrow \text{Ca(OH)}_2\text{(aq)}$; [1 mark]
- All state symbols correct, $\text{Ca(OH)}_2\text{(aq)}$, $\text{CO}_2\text{(g)}$, $\text{CaCO}_3\text{(s)}$, $\text{H}_2\text{O (l)}$, CaO (s) ; [1 mark]

[Total: 4 marks]

- You are expected to know the formulae of calcium hydroxide, calcium carbonate and calcium oxide
- From Fig. 4.1, you have sufficient information to build the unbalanced reaction equations and deduce the balanced chemical equations
 - Calcium hydroxide
 - $\text{Ca(OH)}_2 \rightarrow \text{CaCO}_3$
 - Calcium hydroxide is also known as limewater, which tests for the presence of CO_2
 $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3$
 - The unbalanced equation is missing one oxygen and two hydrogen atoms on the right hand side, which is a water molecule

- **Remember:** Reducing agents cause reduction in other chemicals and are themselves oxidised
- The oxidation number of the magnesium starts at 0 in Mg and becomes +2 in MgCl_2
- This increase in the oxidation number is oxidation, which means that magnesium is the reducing agent
- For part (iii)
 - Magnesium sulfate is soluble in water
 - Titanium metal is solid and does not dissolve in / react with water
 - This means that the titanium can be filtered from the mixture

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- The delocalised electrons (in strontium) are closer to the positive ions / cations / atom / nuclei

OR

The positive ions / cations / atoms (in strontium) are smaller

OR

The positive ions / cations / atoms have fewer (electron) shells / levels; [1 mark]

- So, strontium has a stronger attraction between the positive ions / cations / atoms / nuclei and the delocalised electrons

OR

So, strontium has stronger metallic bonding; [1 mark]

[Total: 2 marks]

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 - The elements have the same equivalent nuclear charge of +2
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- $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$; [1 mark]
- $\text{CaO}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{Ca(OH)}_2(\text{aq})$; [1 mark]
- All state symbols correct, $\text{Ca(OH)}_2(\text{aq})$, $\text{CO}_2(\text{g})$, $\text{CaCO}_3(\text{s})$, $\text{H}_2\text{O}(\text{l})$, $\text{CaO}(\text{s})$; [1 mark]

[Total: 4 marks]

- You are expected to know the formulae of calcium hydroxide, calcium carbonate and calcium oxide
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 - Calcium hydroxide
 - $\text{Ca(OH)}_2 \rightarrow \text{CaCO}_3$
 - Calcium hydroxide is also known as limewater, which tests for the presence of CO_2
 $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3$
 - The unbalanced equation is missing one oxygen and two hydrogen atoms on the right-hand side, which is a water molecule
 $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$
 - Calcium carbonate
 - $\text{CaCO}_3 \rightarrow \text{CaO}$
 - This unbalanced equation is missing one carbon and two oxygen atoms on the right-hand side, which is a carbon dioxide molecule
 $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
 - Calcium oxide
 - $\text{CaO} \rightarrow \text{Ca(OH)}_2$
 - This unbalanced equation is, again, missing one oxygen and two hydrogen atoms on the right-hand side, which is a water molecule
 $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$
- **Remember:** The question asks for state symbols so you need to add state symbols to all of the equations
 - It is more common to be asked to include state symbols for one of the specific equations but you should be adding state symbols as part of your good practice

4b

b)

i) How calcium hydroxide increases the pH of acidic soil:

- The solid calcium hydroxide dissolves (in rainwater / the moisture of the soil); [1 mark]
- This releases hydroxide ions / OH^- into the soil; [1 mark]
- The hydroxide ions / OH^- react with / remove / neutralise the (acidic) hydrogen ions / H^+ ; [1 mark]

ii) The **ionic** equation to show the reaction that causes the pH of the soil to increase is:

- $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$; [1 mark]

[Total: 4 marks]

- Questions like this often see extra, unnecessary and incorrect answers
 - This is because part (i) of the question asks specifically about calcium hydroxide but the introductory sentence talks about calcium carbonate, calcium hydroxide and calcium oxide
- **Careful:** The calcium hydroxide is being added as a solid to a field
 - This means that you need to explain how the solid calcium hydroxide releases the hydroxide ions
 - The hydroxide ions will then react with the hydrogen ions in the soil
 - As more hydrogen ions are removed:
 - The soil becomes less acidic
 - The soil becomes more alkaline
 - This means that the pH of the soil increases
- The increase in the pH is a result of a neutralisation reaction, so you should write the ionic equation for neutralisation as your answer to part (ii)
 - $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$

4c

c)

i) The chemical reaction of calcination is:

- Thermal decomposition; [1 mark]

ii) Strontium carbonate has greater thermal stability than calcium carbonate because:

- Strontium ion has a larger / greater ionic radius

OR

The strontium ion / cation has a lower / decreased charge density; [1 mark]

- So, the polarising / distorting power of the strontium ion / cation decreases; [1 mark]
- Which means that there is less polarisation / distortion of (the electron cloud) of the carbonate ion / anion; [1 mark]

[Total: 4 marks]

- You should be aware that heating calcium carbonate results in its decomposition to calcium oxide and carbon dioxide
- The thermal stability of the Group 2 carbonates increases as you move down the group
 - The smaller positive metal ions at the top of the group polarise the anions more than the larger ions at the bottom of the group
 - These small positive metal ions attract the delocalised electrons in the carbonate ion towards themselves
 - The higher the charge and the smaller the ion; the higher the polarising power
 - The more polarised the carbonate ions are; the more likely they are to thermally decompose as the covalent bonds within them are weaker

4d

d)

i) The coloured gas and its colour are:

- Nitrogen dioxide / nitrogen(IV) oxide / NO_2 **OR** dinitrogen tetroxide / N_2O_4
AND
Brown; [1 mark]

ii) The identity and test, including the result, for the colourless gas are:

- Oxygen / O_2
AND
Relights a glowing splint; [1 mark]

[Total: 2 marks]

- The hydrated calcium nitrate is gently heated to remove the water of crystallisation
- Further, heating then causes the calcium nitrate to decompose into calcium oxide, nitrogen dioxide and oxygen
 - $2\text{Ca}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{CaO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$
 - Nitrogen dioxide is a dark brown gas
 - Some mark schemes may allow the use of red-brown or reddish-brown as the colour
 - Most mark schemes will not allow red as the colour
 - Oxygen is the colourless gas
 - You should know the gas test for oxygen as well as carbon dioxide, hydrogen, ammonia and chlorine

5a

a) The first ionisation energy of calcium (590 kJ mol^{-1}) is greater than that of strontium

thermally decompose as the covalent bonds within them are weaker

4d

d)

i) The coloured gas and its colour are:

- Nitrogen dioxide / nitrogen(IV) oxide / NO_2 **OR** dinitrogen tetroxide / N_2O_4
AND
Brown; [1 mark]

ii) The identity and test, including the result, for the colourless gas are:

- Oxygen / O_2
AND
Relights a glowing splint; [1 mark]

[Total: 2 marks]

- The hydrated calcium nitrate is gently heated to remove the water of crystallisation
- Further, heating then causes the calcium nitrate to decompose into calcium oxide, nitrogen dioxide and oxygen
 - $2\text{Ca}(\text{NO}_3)_2 (\text{s}) \rightarrow 2\text{CaO} (\text{s}) + 4\text{NO}_2 (\text{g}) + \text{O}_2 (\text{g})$
 - Nitrogen dioxide is a dark brown gas
 - Some mark schemes may allow the use of red-brown or reddish-brown as the colour
 - Most mark schemes will not allow red as the colour
 - Oxygen is the colourless gas
 - You should know the gas test for oxygen as well as carbon dioxide, hydrogen, ammonia and chlorine

5a

a) The first ionisation energy of calcium (590 kJ mol^{-1}) is greater than that of strontium (550 kJ mol^{-1}) because:

- Calciums outer / valence electrons are closer to the nucleus
OR
Calcium has a smaller atomic radius

OR

Calcium has less electrons shells; [1 mark]

- So, calcium has less shielding

AND

Which means that more energy is required to overcome the stronger force of attraction (between the outer / valence electron and the nucleus); [1 mark]

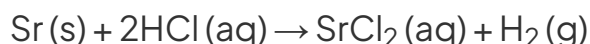
[Total: 2 marks]

- Answers to questions about ionisation energies should include:
 - The distance of the outer electrons from the nucleus
 - The amount of shielding that the outer electrons have from the nucleus
 - The electrostatic force of attraction of the outer electrons to the nucleus
 - The amount of energy required to overcome these electrostatic forces of attraction

5b

b)

i) The balanced chemical equation, including state symbols, to show the formation of strontium chloride from strontium and hydrochloric acid is:



- Correct chemical species

AND

Correct balancing; [1 mark]

- Correct state symbols; [1 mark]

ii) This reaction is not used to produce strontium chloride on an industrial scale because:

- It is a vigorous / violent reaction

OR

There is a risk of explosion; [1 mark]

[Total: 3 marks]

- **Remember:** M.A.S.H - Metal + Acid → Salt + Hydrogen
 - Due to the formulae of strontium chloride and hydrogen, this is one of the nicer

equations to balance

- The reaction between magnesium and hydrochloric acid is quite vigorous
 - The reactivity of the metals increases as you move down the group
 - This means that the reaction of strontium with hydrochloric acid is vigorous and violent
 - On an industrial scale, this would mean that there is the risk of an explosion

5c

c) Drinking magnesium sulfate solution is an effective treatment for barium poisoning because:

- Insoluble barium sulfate is formed

OR

Barium ions are removed (as a precipitate); [1 mark]

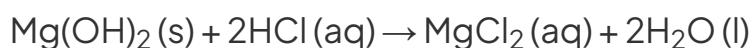
- $\text{Ba}^{2+} + \text{SO}_4^{2-} \rightarrow \text{BaSO}_4$; [1 mark]

[Total: 2 marks]

- This question is a test of your knowledge of the solubility of the group 2 sulfates
- The solubility of the group 2 sulfates decreases as you move down the group
- Magnesium sulfate is soluble in water (hence being able to make the solution)
- Barium sulfate is insoluble, which means that it will pass through the body after it has been formed

5d

d) The remedy that might cause the person taking it to have wind is:



- Calcium carbonate

AND

(Because,) the reaction of calcium carbonate with hydrochloric acid forms carbon dioxide (gas); [1 mark]

- Both reactions form the metal chloride and water; [1 mark]

[Total: 2 marks]

- Although they are not necessary, it can be helpful to write both chemical equations:
 - $\text{Mg}(\text{OH})_2 + 2\text{HCl} \rightarrow \text{MgCl}_2 + 2\text{H}_2\text{O}$
 - $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$
- **Careful:** This question asks you to compare the reactions **AND** give a reason for your choice of the remedy that might cause wind
 - The comparison of the equations shows that both reactions for a group 2 metal chloride and water
 - The difference is that calcium carbonate also forms carbon dioxide
 - Carbon dioxide is a gas and could potentially cause wind in the person taking it

